

SMIT VELANI

Boston, MA | +1 857-426-7025 | velani.sm@northeastern.edu | LinkedIn | GitHub | Portfolio

Education

Northeastern University, Boston, MA

2026 – 2028

Master of Science in Data Science

GPA: 4.0/4.0

Adani Institute of Infrastructure Engineering (GTU)

2021 – 2025

B.E. Information Communication & Technology

CGPA: 8.24/10

Technical Skills

Languages: Python, SQL, JavaScript, PHP, HTML, CSS, SwiftUI

Libraries: Pandas, NumPy, Matplotlib, Scikit-Learn, TensorFlow, PyTorch, OpenCV, Flask, FastAPI, LangChain, D3.js

Tools & Databases: PostgreSQL, MySQL, MongoDB, FAISS, MLflow, Power BI, Git, GitHub, AWS, VS Code

Machine Learning: CNN, Transfer Learning, RAG, LLM Integration, AutoML, SHAP, K-Means, ARIMA, Anomaly Detection

Work Experience

Unified Mentor Pvt. Ltd. | Data Analyst Intern Certificate

Jan 2025 – Apr 2025

Stock Market Data Analysis System GitHub

Python | Pandas | NumPy | Matplotlib | Seaborn | yfinance | Tkinter

- Integrated **Yahoo Finance API** to fetch and cache real-time **OHLCV** data for **4+ stocks**, automating multi-stage cleaning, deduplication and validation using **Pandas** across **250+ daily records**.
- Engineered **11 technical indicators** from scratch (SMA, EMA, Bollinger Bands, RSI, MACD) using fully vectorised **NumPy** operations with no external TA library, ensuring consistent performance across all timeframes.
- Computed **Sharpe Ratio**, **Max Drawdown** and annualised return metrics for quantitative risk-return profiling, enabling side-by-side portfolio performance comparison across all tracked stocks.
- Constructed an interactive **Tkinter dashboard** with **20 clickable charts**, real-time chart switching and an automated data validation pipeline with alert flags for anomalous readings.

IBM SkillsBuild | Data Analytics Intern (CSRBOX) Certificate

Jun 2024 – Aug 2024

Plant Health Prediction System GitHub

Python | NumPy | Flask | TensorFlow | OpenCV | Keras

- Deployed a **MobileNetV2 Transfer Learning** CNN achieving **89% diagnostic accuracy** across **15 disease classes** and **3 plant species** (Pepper, Potato, Tomato) on the **PlantVillage** benchmark dataset.
- Built an **OpenCV** preprocessing pipeline (denoising, HSV-based lesion analysis, Canny edge detection) and implemented **two-phase training** with EarlyStopping and ReduceLROnPlateau callbacks for stable convergence.
- Applied **data augmentation** strategies (rotation, flipping, zoom) to improve model generalisation, then shipped an end-to-end **Flask** web app with drag-and-drop interface for real-time CNN inference.

Projects

Intelligent Data Platform (IDP) GitHub

AutoML | MLOps | Full-Stack Project

Python | FastAPI | React | XGBoost | SHAP | Groq LLaMA | MLflow

- Architected a full-stack **AutoML platform** (FastAPI, React) that automatically cleans any uploaded CSV, detects the problem type (classification, regression, or clustering), trains multiple models and selects the best one.
- Engineered **cost-aware model selection** using a configurable business cost matrix with a **recall-floor** guardrail, leakage-free **SMOTE** cross-validation folds and adaptive **SHAP** explainers across tree, linear and kernel model types.
- Achieved **AUC-ROC 0.981** and **AUC-PR 0.848** on a **284K-row** imbalanced fraud dataset; integrated **PSI** drift detection, 14 pytest tests with **GitHub Actions CI** and **Groq LLaMA** PDF report generation.

Global Economic Intelligence Dashboard GitHub

Full-Stack AI | Data Science Project

Python | Flask | D3.js | Scikit-Learn | Groq LLaMA | ARIMA

- Integrated **World Bank API** across **190+ countries** with **ARIMA**-based 5-year GDP forecasting, **Isolation Forest** anomaly detection and **VADER** sentiment analysis on live news headlines via NewsAPI.
- Rendered an interactive **D3.js** choropleth map with a 4-metric toggle (GDP, inflation, unemployment, per capita), **K-Means** peer clustering and an animated **GDP bar chart race** spanning 2000 to 2024.
- Developed a **recession risk predictor** scoring countries 0-100% using GDP growth, inflation and unemployment, and integrated **Groq LLaMA 3.3** AI summaries with **WebSocket** updates via **Flask-SocketIO**.

DailyBrief AI GitHub

AI Pipeline | Deployed Project

Python | Groq LLaMA | XGBoost | Gmail API | MongoDB Atlas | Railway

- Designed a **6-layer AI pipeline** that ingests **Gmail** and **Google Calendar** APIs, parses inbox data and schedules, and automatically generates a personalized daily intelligence brief three times per day.
- Parsed raw email content using **spaCy** and **VADER** to detect deadlines and urgency; trained an **XGBoost** priority classifier with a self-improving feedback loop persisted to **MongoDB Atlas**.
- Shipped **Groq LLaMA 3.3** brief generation with **APScheduler** cron jobs delivering formatted HTML emails at 8:30 AM, 2:00 PM and 8:00 PM daily via **Resend API** on **Railway** cloud 24/7.